Trevor Teolis

Postdoctoral Researcher, Rice University
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Research Interests

Partial differential equations, kinetic theory, interacting particle systems, collective dynamics, sampling methods, and machine learning.

Education

University of Illinois at Chicago, Chicago, IL
Ph.D., Mathematics, April 2025.
Dissertation: Well-posedness, long-time behavior, and multi-scale description of models of collective dynamics.
Advisor: Roman Shvydkoy.

University of Maryland, College Park, College Park, MD
B.S., Mathematics and Computer Science, May 2019.

Research Experience

Rice University, Houston, TX
Postdoctoral Researcher, Aug 2025 – present.
Advisor: Maarten de Hoop. Studying convergence of sampling methods and foundation models through a Boltzmann framework for transformers, viewed as interacting particle systems.

University of Illinois at Chicago, Chicago, IL
Doctoral Researcher, Aug 2020 – May 2025.
Developed a general theory for a new alignment model, establishing well-posedness, long-time behavior, and rigorous multi-scale analysis, supported by numerical experiments. Analyzed the long-time behavior of coupled Fokker–Planck–Navier–Stokes systems describing interacting particles in a fluid.

University of Maryland, College Park, College Park, MD
Undergraduate Researcher, May 2021 – Aug 2021.
Advisor: Peter Nandori. Proved convergence of random-walk particle models on lattices with absorbing boundaries to the heat equation, with extensions to Sinai billiards.

Publications

1. R. Shvydkoy, T. Teolis, "Microscopic, mesoscopic, and macroscopic descriptions of the Euler Alignment System with adaptive communication strength," *Discrete and Continuous Dynamical Systems*, accepted (2025).

2. R. Shvydkoy, T. Teolis, "Well-posedness and long-time behavior of the Euler Alignment System with adaptive communication strength," *Abel Symposium Proceedings*, accepted (2024).

3. P. Nandori, T. Teolis, "Local Equilibrium of Particle Density in Planar Lorentz Processes," *Nonlinearity*, 2021.

Preprints

1. R. Shvydkoy, T. Teolis, "Long-time behavior of the Fokker–Planck–Navier–Stokes Alignment System." arXiv:2508.07415.

2. A. Chertock, R. Shvydkoy, T. Teolis, "Modulation of the monokinetic limit for models of collective dynamics." arXiv:2508.05478.

Awards

- Victor Twersky Memorial Award, University of Illinois at Chicago, 2024.
- Award for Graduate Research, University of Illinois at Chicago, 2024.
- Travel Award, University of Illinois at Chicago, 2024.

Talks and Presentations

Invited Talks

- "Unconditional alignment of a Fokker–Planck–Navier–Stokes system of interacting particles," *Models of Emergence and Collective Dynamics*, 15th AIMS Conference, Athens, Greece, July 2026. (upcoming)
- "Exponential convergence for second-order SVGD," *Measure Flows for Inverse Problems and Machine Learning*, SIAM UQ, Minneapolis, March 2026. (upcoming)

Contributed Talks

- "Interacting particle systems in machine learning and inference," GMIG Annual Workshop, Rice University, Sept 2025.
- "The s-model and unconditional alignment in a fluid," *Emergent Macroscopic Phenomena in Non-equilibrium Statistical Mechanics*, GSSI, Italy, May 2025.
- "A new model of collective dynamics with adaptive communication strength," *KinMAT Workshop*, University of Warsaw, June 2024.
- "Proof techniques for alignment and flocking," IDEAL Seminar (online), March 2024.

Poster Sessions

- "The s-model and unconditional alignment in a fluid," *Workshop on Kinetic Theory and Fluids*, UW–Madison, March 2025.
- "The s-model and unconditional alignment in a fluid," *Emergent Behavior in Complex Systems*, IMSI, Chicago, March 2025.

Departmental and Group Talks

- "Long-time behavior and multi-scale description of models of collective dynamics," Ph.D. Defense, UIC, April 2025.
- "Sampling with second-order SVGD and in Bayes–Hilbert space," GMIG Weekly Meeting, Rice University, Oct 2025.
- "A new model of collective dynamics with adaptive strength," Analysis and Applied Math Seminar, UIC, Feb 2024.

- “Review of ‘Transformers are Universal In-Context Learners’,” Transformers Reading Group, Dec 2024.

Workshops and Summer Schools

- Emergent Macroscopic Phenomena, GSSI, Italy, May 2025.
- Workshop on Kinetic Theory and Fluids, UW–Madison, March 2025.
- Emergent Behavior in Complex Systems, IMSI, Chicago, March 2025.
- Winter School: Boundary and Singularity in Fluid Mechanics, Stony Brook, Jan 2025.
- KinMAT Workshop, University of Warsaw, June 2024.
- Summer School: Fluid Dynamics, UMD, July 2023.
- Summer School: PDEs, Duke, Aug 2022.

Teaching

University of Illinois at Chicago
Teaching Assistant for Calculus I–III, Differential Equations, and Linear Algebra. Led discussion sections, mini-lectures, and office hours. Received strong student evaluations highlighting clarity and enthusiasm.

Professional Experience

Rampart Communications, Linthicum, MD
Summer Doctoral Intern, Jun – Aug 2024. Developed and implemented polar-coded modulation schemes.

Argonne National Laboratory, Lemont, IL
NSF Summer Intern, Jun – Aug 2021. Developed algorithms for mixed-integer PDE-constrained optimization.

National Institutes of Health, Bethesda, MD
Undergraduate Researcher and Software Engineer, Jan 2018 – Jul 2020. Built finite-element models of brain tumor growth using MRI and DTI data.

References available upon request.
Updated October 2025